

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I C RUTHIKA REDDY(192421138) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Answer:

Grey level transformation is an image enhancement technique used to modify the intensity values of pixels in an image. Its main purpose is to improve image quality by enhancing contrast, highlighting important features, and making objects more visible for better analysis and interpretation.

(b) Apply a negative transformation ($s = 255 - r$) to the pixel values ($r = [50, 100, 150, 200]$). Compute the transformed values.

Given:

- Formula: ($s = 255 - r$)
- Pixel values: ($r = [50, 100, 150, 200]$)

Calculation:

Original Pixel (r) Transformed Pixel ($s = 255 - r$)

50 205

100 155

150 105

200 55

Final Answer: [205, 155, 105, 55]

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Answer:

Log transformation expands low-intensity (dark) pixel values while compressing high-intensity (bright) pixel values. As a result, details hidden in darker regions become more visible, improving the overall image quality.

(b) Given ($s = c \log(1+r)$), where ($c = 10$), compute (s) for ($r = [0, 10, 50, 100]$).

Given:

- Formula: ($s = 10 \log(1+r)$)

- Using natural logarithm ($\ln$)

Calculation:

r Calculation s

0 $(10 \ln(1))$ 0.00

10 $(10 \ln(11))$ 23.98

50 $(10 \ln(51))$ 39.32

100 $(10 \ln(101))$ 46.15

Final Answer: [0.00, 23.98, 39.32, 46.15]

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Answer:

Histogram equalization improves image contrast by redistributing pixel intensity values across the available intensity range. This enhances the visibility of details in both dark and bright regions of an image.

(b) Given a 3-bit image with the following histogram, compute the cumulative distribution and equalized intensity values.

Intensity Frequency

0 2

1 2

2 4

3 8

Step 1: Total Number of Pixels

$$N = 2 + 2 + 4 + 8 = 16$$

For a 3-bit image:

$$L = 2^3 = 8$$

Step 2: Compute Cumulative Distribution Function (CDF)

Intensity Frequency CDF

0	2	2
1	2	4
2	4	8
3	8	16

Step 3: Compute Equalized Intensities

Formula:

$$s = (L - 1) \times (CDF / N)$$

Intensity CDF Equalized Intensity

0	2	1
1	4	2
2	8	4
3	16	7

Final Answer:

Intensity CDF Equalized Intensity

0	2	1
1	4	2
2	8	4
3	16	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Answer:

Spatial filtering for smoothing reduces image noise and small intensity variations by averaging neighboring pixel values, producing a smoother image.

(b) Apply a 3×3 averaging filter to the following region. Compute the new center pixel value.

Given matrix:

```
10 20 30
20 30 40
30 40 50
```

Step 1: Sum of all pixels

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Step 2: Compute Average

$$270 \div 9 = 30$$

Final Answer: 30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Answer:

Sharpening enhances image details by emphasizing edges and fine structures. It increases the contrast between neighboring pixels, making object boundaries and important features clearer.

(b) Apply the Laplacian mask to the given matrix and compute the output at the center.

Laplacian Mask

```
0 -1 0
-1 4 -1
0 -1 0
```

Image Matrix

10 10 10

10 20 10

10 10 10

Calculation

$$(4 \times 20) - (10 + 10 + 10 + 10)$$

$$= 80 - 40$$

$$= 40$$

Final Answer: 40